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Chapter 29 - Classical Orthogonal Polynomials

The power-series and Frobenius methods do more than produce infinite series. For certain differential equations and certain parameter values, a recurrence relation forces the series to stop. The result is a polynomial solution.

Four important families arise this way:

- Legendre polynomials
- Chebyshev polynomials
- Hermite polynomials
- Laguerre polynomials

These families are useful because they combine three structures: each member solves a differential equation, neighbouring members satisfy recurrence relations, and different members are orthogonal under an appropriate weighted integral.

This chapter will:

- explain how a parameter turns an infinite series into a polynomial
- derive the coefficient recurrence for each classical equation
- state the normalization used for every polynomial family
- connect differential equations, Rodrigues formulas, and generating functions
- derive weighted orthogonality from self-adjoint form
- calculate expansion coefficients using weighted inner products
- build a reliable workflow for recognising and checking classical polynomials

Why Polynomial Solutions Appear

The Goal Of This Block

The goal is to understand what makes a classical equation special and why its parameter is restricted to a nonnegative integer.

> **the equation does not begin by assuming a polynomial; its recurrence reveals when a polynomial is possible.**

A Parameter-Dependent Equation

Consider a family of equations written schematically as:

$$A(x)y'' + B(x)y' + \lambda y = 0.$$

Here:

- x is the independent variable
- $y(x)$ is the unknown function
- λ is a parameter
- $A(x)$ and $B(x)$ are fixed coefficient functions

Different values of λ usually produce different solutions. At special values, however, the power-series recurrence can contain a zero numerator. Every later coefficient in one parity chain then becomes zero.

The key mechanism is:

$$\boxed{\text{special parameter} \longrightarrow \text{zero recurrence factor} \longrightarrow \text{series termination} \longrightarrow \text{polynomial}}$$

The Four Equations

For $n = 0, 1, 2, \ldots$, the four equations used in this chapter are:

$$\begin{aligned} \text{Legendre:} \quad & (1-x^2)y'' - 2xy' + n(n+1)y = 0, \\ \text{Chebyshev:} \quad & (1-x^2)y'' - xy' + n^2 y = 0, \\ \text{Hermite:} \quad & y'' - 2xy' + 2ny = 0, \\ \text{Laguerre:} \quad & xy'' + (1-x)y' + ny = 0. \end{aligned}$$

The symbol n labels the polynomial degree. It is not a summation index in these equations. When a power series is introduced, a different letter such as k will be used for the series index.

Normalization Is Part Of The Name

A homogeneous equation does not distinguish y from a nonzero multiple Cy. If y solves the equation, then Cy also solves it. A normalization selects one representative and gives the named polynomial a definite scale.

We will use:

$$\begin{aligned} & P_n(1) = 1 && \text{for Legendre polynomials,} \\ & T_n(1) = 1 && \text{for Chebyshev polynomials,} \\ & H_n(x) = 2^n x^n + \text{lower powers} && \text{for physicists' Hermite polynomials,} \\ & L_n(0) = 1 && \text{for Laguerre polynomials.} \end{aligned}$$

For example, $1 - 2x + x^2/2$ and $2 - 4x + x^2$ solve the same $n = 2$ Laguerre equation. The first is the standard normalized polynomial $L_2(x)$; the second is $2L_2(x)$.

The practical meaning:

solving the differential equation determines a polynomial only up to scale; the normalization determines its conventional name.

First Members Of The Four Families

With the normalizations above:

$$\begin{aligned}
&P_0(x) = 1, \quad P_1(x) = x, \quad P_2(x) = \frac{1}{2}(3x^2 - 1), \quad P_3(x) = \frac{1}{2}(5x^3 - 3x), \\
&T_0(x) = 1, \quad T_1(x) = x, \quad T_2(x) = 2x^2 - 1, \quad T_3(x) = 4x^3 - 3x, \\
&H_0(x) = 1, \quad H_1(x) = 2x, \quad H_2(x) = 4x^2 - 2, \quad H_3(x) = 8x^3 - 12x, \\
&L_0(x) = 1, \quad L_1(x) = 1 - x, \quad L_2(x) = 1 - 2x + \frac{1}{2}x^2, \quad L_3(x) = 1 - 3x + \frac{3}{2}x^2 - \frac{1}{6}x^3.
\end{aligned}$$

Do not try to memorise this whole display. The methods developed below rebuild these polynomials from their equations and normalizations.

Block 1 Summary

Each classical family is a sequence of normalized polynomial solutions. The integer n chooses the equation and, after normalization, identifies the degree-n polynomial.

Legendre Polynomials From A Power Series

The Goal Of This Block

The goal is to derive the Legendre recurrence, see exactly why one coefficient chain terminates, and construct $P_3(x)$ without guessing it.

Begin With The Legendre Equation

For a fixed nonnegative integer n, consider:

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0.$$

The coefficient $1-x^2$ vanishes at $x = \pm 1$. The origin is an ordinary point, so seek a power series centred at 0:

$$y = \sum_{k=0}^{\infty} a_k x^k.$$

Differentiate and reindex:

$$y' = \sum_{k=0}^{\infty} (k+1)a_{k+1}x^k,$$
$$y'' = \sum_{k=0}^{\infty} (k+2)(k+1)a_{k+2}x^k.$$

Substitute Every Term

Start by expanding the equation:

$$y'' - x^2y'' - 2xy' + n(n+1)y = 0.$$

The first term is already indexed by x^k:

$$y'' = \sum_{k=0}^{\infty} (k+2)(k+1)a_{k+2}x^k.$$

For the product x^2y'', use the unreindexed derivative and multiply by x^2:

$$\begin{aligned} x^2y'' &= x^2 \sum_{k=2}^{\infty} k(k-1)a_k x^{k-2} \\ &= \sum_{k=2}^{\infty} k(k-1)a_k x^k. \end{aligned}$$

Similarly:

$$\begin{aligned} 2xy' &= 2x \sum_{k=1}^{\infty} k a_k x^{k-1} \\ &= \sum_{k=1}^{\infty} 2k a_k x^k. \end{aligned}$$

Finally:

$$n(n+1)y = \sum_{k=0}^{\infty} n(n+1)a_k x^k.$$

The factors $k(k-1)$ and k vanish automatically at the omitted starting indices, so all four sums may be compared from $k=0$ onward.

Derive The Legendre Recurrence

The coefficient of x^k is:

$$(k+2)(k+1)a_{k+2} - k(k-1)a_k - 2ka_k + n(n+1)a_k.$$

Combine the three terms multiplying a_k:

$$\begin{aligned} & -k(k-1) - 2k + n(n+1) \\ &= -k^2 - k + n(n+1) \\ &= n(n+1) - k(k+1). \end{aligned}$$

Hence:

$$(k+2)(k+1)a_{k+2} + [n(n+1) - k(k+1)]a_k = 0.$$

Subtract the a_k term and divide by $(k+2)(k+1)$:

$$\boxed{a_{k+2} = \frac{k(k+1) - n(n+1)}{(k+2)(k+1)} a_k}.$$

Factor the numerator:

$$\begin{aligned} k(k+1) - n(n+1) &= k^2 + k - n^2 - n \\ &= (k-n)(k+n+1). \end{aligned}$$

Therefore the equivalent factored recurrence is:

$$\boxed{a_{k+2} = \frac{(k-n)(k+n+1)}{(k+2)(k+1)} a_k}.$$

Why The Series Terminates

The recurrence advances by two indices:

$$a_0 \to a_2 \to a_4 \to \cdots, \qquad a_1 \to a_3 \to a_5 \to \cdots.$$

These are the even and odd coefficient chains. When $k = n$, the recurrence numerator contains:

$$k - n = n - n = 0.$$

Consequently:

$$a_{n+2} = 0.$$

The same recurrence then gives $a_{n+4} = a_{n+6} = \cdots = 0$. The chain having the same parity as n terminates at degree n.

The other parity chain generally continues forever. To obtain the Legendre polynomial, set the starting coefficient of that nonterminating chain to zero.

Worked Construction Of P Three

Set $n = 3$. The recurrence becomes:

$$a_{k+2} = \frac{(k-3)(k+4)}{(k+2)(k+1)} a_k.$$

Because $n = 3$ is odd, use the odd chain and set $a_0 = 0$. Begin with arbitrary a_1.

For $k = 1$:

$$\begin{aligned} a_3 &= \frac{(1-3)(1+4)}{(3)(2)} a_1 \\ &= \frac{(-2)(5)}{6} a_1 \\ &= -\frac{5}{3} a_1. \end{aligned}$$

For $k = 3$:

$$\begin{aligned} a_5 &= \frac{(3-3)(3+4)}{(5)(4)} a_3 \\ &= 0. \end{aligned}$$

Thus the terminating odd solution is:

$$y(x) = a_1 x - \frac{5}{3} a_1 x^3.$$

Factor out a_1:

$$y(x) = a_1 \left(x - \frac{5}{3} x^3 \right).$$

Now apply the normalization $P_3(1) = 1$:

$$\begin{aligned} 1 &= a_1\left(1 - \frac{5}{3}\right) \\ &= -\frac{2}{3}a_1. \end{aligned}$$

From $1 = -2a_1/3$, multiply both sides by $-3/2$:

$$a_1 = -\frac{3}{2}.$$

Substitute this value into the polynomial:

$$\boxed{P_3(x) = \frac{1}{2}(5x^3 - 3x)}.$$

Verify The Constructed Polynomial

For $n = 3$, the Legendre equation is:

$$(1 - x^2)y'' - 2xy' + 12y = 0.$$

Using $P_3 = (5x^3 - 3x)/2$:

$$P_3' = \frac{1}{2}(15x^2 - 3), \qquad P_3'' = 15x.$$

Substitute these derivatives:

$$\begin{aligned} &(1 - x^2)P_3'' - 2xP_3' + 12P_3 \\ &= (1 - x^2)(15x) - 2x\left(\frac{15x^2 - 3}{2}\right) + 12\left(\frac{5x^3 - 3x}{2}\right) \\ &= (15x - 15x^3) + (-15x^3 + 3x) + (30x^3 - 18x) \\ &= 0. \end{aligned}$$

The normalization is also immediate:

$$P_3(1) = \frac{1}{2}(5-3) = 1.$$

Block 2 Summary

The Legendre recurrence links coefficients two places apart. At $k = n$, its numerator vanishes, so the parity chain matching n terminates. The condition $P_n(1) = 1$ then fixes the remaining scale.

Chebyshev Polynomials And The Hidden Angle

The Goal Of This Block

The goal is to derive the Chebyshev recurrence and then explain why the substitution $x = \cos\theta$ turns the equation into simple harmonic motion.

Derive The Series Recurrence

The Chebyshev equation of degree n is:

$$(1-x^2)y'' - xy' + n^2 y = 0.$$

Use:

$$y = \sum_{k=0}^{\infty} a_k x^k.$$

As in the Legendre calculation:

$$\begin{aligned} y'' &= \sum_{k=0}^{\infty} (k+2)(k+1)a_{k+2}x^k, \\ x^2 y'' &= \sum_{k=0}^{\infty} k(k-1)a_k x^k, \\ xy' &= \sum_{k=0}^{\infty} k a_k x^k. \end{aligned}$$

Substitute these expressions into:

$$y'' - x^2 y'' - xy' + n^2 y = 0.$$

The coefficient of x^k is:

$$(k+2)(k+1)a_{k+2} - k(k-1)a_k - ka_k + n^2 a_k.$$

Combine the middle terms:

$$-k(k-1) - k = -k^2.$$

After using $-k(k-1)-k=-k^2$, the coefficient equation is:

$$(k+2)(k+1)a_{k+2}+(n^2-k^2)a_k=0.$$

Solve for the later coefficient:

$$\boxed{a_{k+2}=\frac{k^2-n^2}{(k+2)(k+1)}a_k=\frac{(k-n)(k+n)}{(k+2)(k+1)}a_k}.$$

At $k=n$, the factor $k-n$ is zero. The coefficient chain with the same parity as n therefore terminates.

Worked Construction Of T Four

Set $n=4$. Since the required polynomial has even degree, choose the even chain and set $a_1=0$. The recurrence is:

$$a_{k+2}=\frac{(k-4)(k+4)}{(k+2)(k+1)}a_k.$$

For $k=0$:

$$\begin{aligned}a_2&=\frac{(0-4)(0+4)}{(2)(1)}a_0\\&=-8a_0.\end{aligned}$$

For $k=2$:

$$\begin{aligned}a_4&=\frac{(2-4)(2+4)}{(4)(3)}a_2\\&=-a_2\\&=8a_0.\end{aligned}$$

For $k = 4$:

$$\begin{aligned} a_6 &= \frac{(4-4)(4+4)}{(6)(5)} a_4 \\ &= 0. \end{aligned}$$

Thus:

$$y(x) = a_0(1 - 8x^2 + 8x^4).$$

At $x = 1$, the bracket equals $1 - 8 + 8 = 1$. The normalization $T_4(1) = 1$ therefore gives $a_0 = 1$:

$$\boxed{T_4(x) = 8x^4 - 8x^2 + 1}.$$

Reveal The Hidden Constant-Coefficient Equation

On $-1 < x < 1$, write:

$$x = \cos\theta$$

and define:

$$Y(\theta) = y(\cos\theta).$$

Differentiate Y with respect to θ. By the chain rule:

$$\begin{aligned} Y' &= y'(x) \frac{dx}{d\theta} \\ &= -y'(x) \sin\theta. \end{aligned}$$

Differentiate once more, using the product rule:

$$\begin{aligned} Y'' &= -\left[y''(x) \frac{dx}{d\theta} \sin\theta + y'(x)\cos\theta \right] \\ &= -\left[-y''(x)\sin^2\theta + y'(x)\cos\theta \right] \\ &= y''(x)\sin^2\theta - y'(x)\cos\theta. \end{aligned}$$

Since $x = \cos\theta$ and $\sin^2\theta = 1 - x^2$:

$$\boxed{Y'' = (1-x^2)y'' - xy'}.$$

The Chebyshev equation consequently becomes:

$$Y'' + n^2 Y = 0.$$

This is a constant-coefficient oscillation equation.

Obtain The Cosine Formula

One solution of $Y'' + n^2 Y = 0$ is:

$$Y(\theta) = \cos(n\theta).$$

Because $x = \cos\theta$, we may take $\theta = \arccos x$. Return to x:

$$\boxed{T_n(x) = \cos(n \arccos x), \qquad -1 \le x \le 1}.$$

This formula automatically satisfies $T_n(1) = 1$, because $\arccos(1) = 0$ and $\cos(0) = 1$.

For $n = 4$, the multiple-angle identity gives:

$$\cos(4\theta) = 8\cos^4\theta - 8\cos^2\theta + 1.$$

Substituting $x = \cos\theta$ recovers:

$$T_4(x) = 8x^4 - 8x^2 + 1.$$

The series calculation and the angle substitution have produced the same normalized polynomial by two independent routes.

Derive The Three-Term Recurrence

The cosine addition identity is:

$$\cos((n+1)\theta) + \cos((n-1)\theta) = 2\cos\theta\cos(n\theta).$$

Use:

$$x = \cos\theta, \qquad T_j(x) = \cos(j\theta).$$

The identity becomes:

$$T_{n+1}(x) + T_{n-1}(x) = 2xT_n(x).$$

Subtract T_{n-1} from both sides:

$$\boxed{T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x)}.$$

For example, using $T_2 = 2x^2 - 1$ and $T_3 = 4x^3 - 3x$:

$$\begin{aligned} T_4 &= 2xT_3 - T_2 \\ &= 2x(4x^3 - 3x) - (2x^2 - 1) \\ &= 8x^4 - 6x^2 - 2x^2 + 1 \\ &= 8x^4 - 8x^2 + 1. \end{aligned}$$

Block 3 Summary

The Chebyshev recurrence terminates at $k = n$. The substitution $x = \cos\theta$ also reveals that the polynomial is a disguised cosine: $T_n(x) = \cos(n \arccos x)$.

Hermite Polynomials And Gaussian Structure

The Goal Of This Block

The goal is to derive the Hermite coefficient recurrence, construct a normalized polynomial, and understand why derivatives of a Gaussian generate the same family.

Derive The Hermite Recurrence

The physicists' Hermite equation is:

$$y'' - 2xy' + 2ny = 0.$$

Use the series:

$$y = \sum_{k=0}^{\infty} a_k x^k.$$

Then:

$$y'' = \sum_{k=0}^{\infty} (k+2)(k+1)a_{k+2}x^k,$$

and:

$$-2xy' = -2\sum_{k=0}^{\infty} k a_k x^k.$$

The final term is:

$$2ny = 2n\sum_{k=0}^{\infty} a_k x^k.$$

Match the coefficient of x^k:

$$(k+2)(k+1)a_{k+2} - 2ka_k + 2na_k = 0.$$

Combine the a_k terms:

$$(k+2)(k+1)a_{k+2} + 2(n-k)a_k = 0.$$

Therefore:

$$\boxed{a_{k+2} = \frac{2(k-n)}{(k+2)(k+1)}a_k}.$$

At $k = n$, the numerator vanishes. The parity chain matching n terminates.

Worked Construction Of H Four

Set $n = 4$ and use the even coefficient chain. The recurrence becomes:

$$a_{k+2} = \frac{2(k-4)}{(k+2)(k+1)}a_k.$$

For $k = 0$:

$$\begin{aligned} a_2 &= \frac{2(0-4)}{(2)(1)}a_0 \\ &= -4a_0. \end{aligned}$$

For $k = 2$:

$$\begin{aligned} a_4 &= \frac{2(2-4)}{(4)(3)}a_2 \\ &= -\frac{1}{3}a_2 \\ &= \frac{4}{3}a_0. \end{aligned}$$

With $a_2 = -4a_0$ and $a_4 = 4a_0/3$, set $k = 4$:

$$a_6 = 0.$$

Thus the terminating solution is:

$$y = a_0\left(1 - 4x^2 + \frac{4}{3}x^4\right).$$

The physicists' normalization requires the leading coefficient to be $2^4 = 16$. Since the leading coefficient above is $4a_0/3$:

$$\frac{4}{3}a_0 = 16.$$

Multiply by $3/4$:

$$a_0 = 12.$$

Therefore:

$$\boxed{H_4(x) = 16x^4 - 48x^2 + 12}.$$

The Rodrigues Formula

Hermite polynomials can also be generated by:

$$\boxed{H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n}\left(e^{-x^2}\right)}.$$

The factor e^{x^2} cancels the Gaussian remaining after repeated differentiation, leaving a polynomial.

For $n = 2$, begin with:

$$\frac{d}{dx} e^{-x^2} = -2x e^{-x^2}.$$

Differentiate again using the product rule:

$$\begin{aligned} \frac{d^2}{dx^2} e^{-x^2} &= -2e^{-x^2} + (-2x)(-2x)e^{-x^2} \\ &= (4x^2 - 2)e^{-x^2}. \end{aligned}$$

Now use $(-1)^2 = 1$ and multiply by e^{x^2}:

$$\begin{aligned} H_2(x) &= e^{x^2}(4x^2 - 2)e^{-x^2} \\ &= 4x^2 - 2. \end{aligned}$$

A Useful Hermite Recurrence

Neighbouring physicists' Hermite polynomials satisfy:

$$\boxed{H_{n+1}(x) = 2xH_n(x) - 2nH_{n-1}(x)}.$$

Use $n = 3$, together with:

$$H_2 = 4x^2 - 2, \qquad H_3 = 8x^3 - 12x.$$

Then:

$$\begin{aligned} H_4 &= 2xH_3 - 6H_2 \\ &= 2x(8x^3 - 12x) - 6(4x^2 - 2) \\ &= 16x^4 - 24x^2 - 24x^2 + 12 \\ &= 16x^4 - 48x^2 + 12. \end{aligned}$$

This agrees with the power-series construction.

Block 4 Summary

The Hermite equation produces a terminating parity chain when n is a nonnegative integer. The standard physicists' normalization fixes the leading coefficient at 2^n, while the Rodrigues formula exposes the family's connection to the Gaussian e^{-x^2}.

Laguerre Polynomials On A Half-Line

The Goal Of This Block

The goal is to derive a one-step recurrence for the Laguerre coefficients and see how its normalization differs from the parity-based families.

Derive The Laguerre Recurrence

The Laguerre equation is:

$$xy'' + (1 - x)y' + ny = 0.$$

Use:

$$y = \sum_{k=0}^{\infty} a_k x^k.$$

First reindex xy''. Starting from the unreindexed second derivative:

$$y'' = \sum_{j=2}^{\infty} j(j-1)a_j x^{j-2},$$

start from $y'' = \sum_{j=2}^{\infty} j(j-1)a_j x^{j-2}$ and multiply by x:

$$xy'' = \sum_{j=2}^{\infty} j(j-1)a_j x^{j-1}.$$

Set $k = j - 1$, so $j = k + 1$:

$$xy'' = \sum_{k=1}^{\infty} (k+1)k a_{k+1} x^k.$$

The remaining derivative terms are:

$$y' = \sum_{k=0}^{\infty} (k+1)a_{k+1}x^k$$

and:

$$-xy' = -\sum_{k=1}^{\infty} ka_k x^k.$$

Finally:

$$ny = n\sum_{k=0}^{\infty} a_k x^k.$$

For $k = 0$, the xy'' and $-xy'$ contributions vanish. For every $k \geq 0$, the coefficient equation can therefore be written as:

$$[(k+1)k + (k+1)]a_{k+1} + (n-k)a_k = 0.$$

Factor the first bracket:

$$(k+1)k + (k+1) = (k+1)^2.$$

Thus:

$$(k+1)^2 a_{k+1} + (n-k)a_k = 0.$$

Solve for a_{k+1}:

$$\boxed{a_{k+1} = \frac{k-n}{(k+1)^2}a_k}.$$

Unlike the two-step recurrences above, this recurrence connects every successive coefficient. At $k = n$, it gives $a_{n+1} = 0$, after which all later coefficients vanish.

Worked Construction Of L Three

Set $n = 3$ and apply the normalization $L_3(0) = 1$. Since $L_3(0) = a_0$, choose:

$$a_0 = 1.$$

For $k = 0$:

$$\begin{aligned} a_1 &= \frac{0-3}{(1)^2} a_0 \\ &= -3. \end{aligned}$$

For $k = 1$:

$$\begin{aligned} a_2 &= \frac{1-3}{(2)^2} a_1 \\ &= -\frac{1}{2}(-3) \\ &= \frac{3}{2}. \end{aligned}$$

For $k = 2$:

$$\begin{aligned} a_3 &= \frac{2-3}{(3)^2} a_2 \\ &= -\frac{1}{9}\left(\frac{3}{2}\right) \\ &= -\frac{1}{6}. \end{aligned}$$

For $k = 3$:

$$a_4 = 0.$$

Using $a_0 = 1$, $a_1 = -3$, $a_2 = 3/2$, and $a_3 = -1/6$, the polynomial is:

$$\boxed{L_3(x) = 1 - 3x + \frac{3}{2}x^2 - \frac{1}{6}x^3}.$$

Verify L Three Directly

For $n = 3$, the Laguerre equation is:

$$xy'' + (1-x)y' + 3y = 0.$$

Differentiate the polynomial:

$$\begin{aligned} L_3' &= -3 + 3x - \frac{1}{2}x^2, \\ L_3'' &= 3 - x. \end{aligned}$$

Substitute one term at a time:

$$xL_3'' = 3x - x^2,$$

$$\begin{aligned} (1-x)L_3' &= (1-x)\left(-3 + 3x - \frac{1}{2}x^2\right) \\ &= -3 + 6x - \frac{7}{2}x^2 + \frac{1}{2}x^3, \end{aligned}$$

and:

$$3L_3 = 3 - 9x + \frac{9}{2}x^2 - \frac{1}{2}x^3.$$

Add the three expressions:

$$\begin{aligned} xL_3'' + (1-x)L_3' + 3L_3 &= (3x - x^2) \\ &\quad + \left(-3 + 6x - \frac{7}{2}x^2 + \frac{1}{2}x^3\right) \\ &\quad + \left(3 - 9x + \frac{9}{2}x^2 - \frac{1}{2}x^3\right) \\ &= 0. \end{aligned}$$

The Laguerre Rodrigues Formula

The normalized Laguerre polynomial is also given by:

$$\boxed{L_n(x) = \frac{e^x}{n!}\frac{d^n}{dx^n}\left(e^{-x}x^n\right)}.$$

For $n = 2$, let:

$$f(x) = e^{-x}x^2.$$

Differentiate once:

$$\begin{aligned} f' &= (-e^{-x})x^2 + e^{-x}(2x) \\ &= e^{-x}(2x - x^2). \end{aligned}$$

Differentiate again:

$$\begin{aligned} f'' &= -e^{-x}(2x - x^2) + e^{-x}(2 - 2x) \\ &= e^{-x}(x^2 - 4x + 2). \end{aligned}$$

Insert this into the Rodrigues formula:

$$\begin{aligned} L_2(x) &= \frac{e^x}{2!}e^{-x}(x^2 - 4x + 2) \\ &= \frac{1}{2}(x^2 - 4x + 2) \\ &= 1 - 2x + \frac{1}{2}x^2. \end{aligned}$$

Block 5 Summary

The Laguerre recurrence advances one coefficient at a time and terminates when $k = n$. The condition $L_n(0) = 1$ sets $a_0 = 1$, while the factor $1/n!$ in the Rodrigues formula produces the same normalization.

Three Ways To Generate A Polynomial Family

The Goal Of This Block

The goal is to distinguish three related tools: differential equations, Rodrigues formulas, and generating functions.

these tools describe the same polynomial families, but they answer different computational questions.

What Each Tool Is For

The **differential equation** explains which functions belong to a family and why special parameter values permit polynomial solutions.

A **Rodrigues formula** constructs one chosen polynomial by repeated differentiation.

A **generating function** packages the whole family into one function of an extra variable t. Expanding in powers of t reveals every polynomial as a coefficient.

These are complementary descriptions. A generating function does not replace the independent variable x; it introduces a bookkeeping variable t.

Rodrigues Formulas And Normalization

Three useful Rodrigues formulas are:

$$P_n(x) = \frac{1}{2^n n!}\frac{d^n}{dx^n}(x^2-1)^n,$$
$$H_n(x) = (-1)^n e^{x^2}\frac{d^n}{dx^n}e^{-x^2},$$
$$L_n(x) = \frac{e^x}{n!}\frac{d^n}{dx^n}(e^{-x}x^n).$$

The prefactors are not decoration. They enforce the normalization convention for the named family.

For example, use the Legendre formula with $n = 2$:

$$P_2(x) = \frac{1}{2^2(2!)}\frac{d^2}{dx^2}(x^2-1)^2.$$

Expand before differentiating:

$$(x^2-1)^2 = x^4 - 2x^2 + 1.$$

Differentiate once:

$$\frac{d}{dx}(x^4 - 2x^2 + 1) = 4x^3 - 4x.$$

Differentiate again:

$$\frac{d^2}{dx^2}(x^2-1)^2 = 12x^2 - 4.$$

Since $2^2(2!) = 8$:

$$P_2(x) = \frac{1}{8}(12x^2 - 4) \\ = \frac{1}{2}(3x^2 - 1).$$

Generating Functions

The following identities define generating functions near $t = 0$:

$$\frac{1}{\sqrt{1 - 2xt + t^2}} = \sum_{n=0}^{\infty} P_n(x)t^n,$$
$$\frac{1 - xt}{1 - 2xt + t^2} = \sum_{n=0}^{\infty} T_n(x)t^n,$$
$$e^{2xt - t^2} = \sum_{n=0}^{\infty} H_n(x)\frac{t^n}{n!},$$
$$\frac{1}{1-t}\exp\left(-\frac{xt}{1-t}\right) = \sum_{n=0}^{\infty} L_n(x)t^n.$$

Read the right sides carefully. The Hermite coefficient of t^n is $H_n(x)/n!$, whereas the other three displays use the polynomial itself as the coefficient.

Worked Hermite Generating-Function Expansion

Use:

$$e^{2xt-t^2} = e^{2xt}e^{-t^2}.$$

Expand each factor through t^3:

$$\begin{aligned} e^{2xt} &= 1 + 2xt + \frac{(2xt)^2}{2!} + \frac{(2xt)^3}{3!} + O(t^4) \\ &= 1 + 2xt + 2x^2t^2 + \frac{4}{3}x^3t^3 + O(t^4), \end{aligned}$$

and:

$$e^{-t^2} = 1 - t^2 + O(t^4).$$

Multiply the truncated series:

$$\begin{aligned} e^{2xt-t^2} &= \left(1 + 2xt + 2x^2t^2 + \frac{4}{3}x^3t^3 + O(t^4)\right)\left(1 - t^2 + O(t^4)\right) \\ &= 1 + 2xt + (2x^2-1)t^2 + \left(\frac{4}{3}x^3 - 2x\right)t^3 + O(t^4). \end{aligned}$$

Compare this with:

$$e^{2xt-t^2} = H_0 + H_1t + H_2\frac{t^2}{2!} + H_3\frac{t^3}{3!} + \cdots.$$

Match coefficients one power at a time:

$$H_0 = 1, \qquad H_1 = 2x.$$

For t^2:

$$\frac{H_2}{2!} = 2x^2 - 1.$$

Multiply by $2! = 2$:

$$H_2 = 4x^2 - 2.$$

For t^3:

$$\frac{H_3}{3!} = \frac{4}{3}x^3 - 2x.$$

Multiply by $3! = 6$:

$$H_3 = 8x^3 - 12x.$$

Neighbour Recurrences

Once two starting polynomials are known, three-term recurrences efficiently generate later members:

$$
\begin{aligned}
(n+1)P_{n+1} &= (2n+1)xP_n - nP_{n-1}, \\
T_{n+1} &= 2xT_n - T_{n-1}, \\
H_{n+1} &= 2xH_n - 2nH_{n-1}, \\
(n+1)L_{n+1} &= (2n+1-x)L_n - nL_{n-1}.
\end{aligned}
$$

For the Legendre family, use $n = 3$ to generate P_4:

$$4P_4 = 7xP_3 - 3P_2.$$

Insert:

$$P_3 = \frac{1}{2}(5x^3 - 3x), \qquad P_2 = \frac{1}{2}(3x^2 - 1).$$

Then:

$$
\begin{aligned}
4P_4 &= 7x\left(\frac{5x^3 - 3x}{2}\right) - 3\left(\frac{3x^2 - 1}{2}\right) \\
&= \frac{1}{2}(35x^4 - 21x^2 - 9x^2 + 3) \\
&= \frac{1}{2}(35x^4 - 30x^2 + 3).
\end{aligned}
$$

Divide by 4:

$$\boxed{P_4(x) = \frac{1}{8}(35x^4 - 30x^2 + 3)}.$$

Common Mistake: Forgetting The Hermite Factorial

From:

$$e^{2xt-t^2} = \sum_{n=0}^{\infty} H_n(x)\frac{t^n}{n!},$$

the coefficient of t^3 is $H_3/3!$, not H_3. If that coefficient is $4x^3/3 - 2x$, the polynomial is:

$$H_3 = 3!\left(\frac{4}{3}x^3 - 2x\right) = 8x^3 - 12x.$$

Block 6 Summary

The differential equation identifies the family, a Rodrigues formula builds a chosen member, a generating function packages all members, and a neighbour recurrence advances efficiently from known lower degrees.

Weighted Orthogonality From The Differential Equation

The Goal Of This Block

The goal is to explain what weighted orthogonality means and derive it from the self-adjoint form of the differential equation.

Weighted Inner Products

On an interval (a, b), define:

$$\boxed{\langle f, g\rangle_w = \int_a^b f(x)g(x)w(x)\,dx}.$$

The function $w(x)$ is the **weight**. Two nonzero functions are orthogonal when:

$$\langle f, g\rangle_w = 0.$$

This is the function-space analogue of perpendicular vectors having zero dot product. The weight changes how strongly different parts of the interval contribute to the comparison.[1]

The Self-Adjoint Pattern

Consider the parameter-dependent equation:

$$\boxed{(p(x)y')' + \lambda w(x)y = 0}.$$

Suppose u solves the equation with parameter λ:

$$(pu')' + \lambda wu = 0,$$

and v solves it with a different parameter μ:

$$(pv')' + \mu wv = 0.$$

[1] In a finite-dimensional dot product, coordinates may be assigned different importance. A weight function performs the continuous version of that reweighting across an interval.

Multiply the first equation by v:

$$v(pu')' + \lambda wuv = 0.$$

For $(pv')' + \mu wv = 0$, multiply by u:

$$u(pv')' + \mu wuv = 0.$$

The first multiplication gave $v(pu')' + \lambda wuv = 0$. Subtract the u-multiplied equation from that result:

$$v(pu')' - u(pv')' + (\lambda - \mu)wuv = 0.$$

The first two terms form a product derivative:

$$v(pu')' - u(pv')' = \frac{d}{dx}\left[p(vu' - uv')\right].$$

Therefore:

$$\frac{d}{dx}\left[p(vu' - uv')\right] + (\lambda - \mu)wuv = 0.$$

Integrate The Identity

Integrate from a to b:

$$\int_a^b \frac{d}{dx}\left[p(vu' - uv')\right] dx + (\lambda - \mu)\int_a^b wuv\, dx = 0.$$

Evaluate the first integral using the Fundamental Theorem of Calculus:

$$[p(vu' - uv')]_a^b + (\lambda - \mu) \int_a^b wuv \, dx = 0.$$

For the classical boundary conditions, the boundary term vanishes. Hence:

$$(\lambda - \mu) \int_a^b wuv \, dx = 0.$$

If $\lambda \neq \mu$, divide by $\lambda - \mu$:

$$\boxed{\int_a^b u(x)v(x)w(x) \, dx = 0}.$$

Thus eigenfunctions belonging to distinct parameter values are orthogonal.

The Four Weights And Intervals

Each classical equation can be written in self-adjoint form:

family	(a, b)	$p(x)$	$w(x)$
P_n	$(-1, 1)$	$1 - x^2$	1
T_n	$(-1, 1)$	$\sqrt{1 - x^2}$	$(1 - x^2)^{-1/2}$
H_n	$(-\infty, \infty)$	e^{-x^2}	e^{-x^2}
L_n	$(0, \infty)$	xe^{-x}	e^{-x}

For example, multiply the Hermite equation by e^{-x^2}:

$$e^{-x^2} y'' - 2xe^{-x^2} y' + 2ne^{-x^2} y = 0.$$

The first two terms combine because:

$$\begin{aligned}\frac{d}{dx}(e^{-x^2}y') &= e^{-x^2}y'' + (e^{-x^2})'y' \\ &= e^{-x^2}y'' - 2xe^{-x^2}y'.\end{aligned}$$

Therefore the self-adjoint Hermite equation is:

$$(e^{-x^2}y')' + 2ne^{-x^2}y = 0.$$

Worked Legendre Orthogonality Check

Verify that P_1 and P_3 are orthogonal on $[-1, 1]$ with weight 1. Use:

$$P_1(x) = x, \qquad P_3(x) = \frac{1}{2}(5x^3 - 3x).$$

Form the weighted inner product:

$$\begin{aligned}\langle P_1, P_3\rangle &= \int_{-1}^{1} x\left[\frac{1}{2}(5x^3 - 3x)\right]dx \\ &= \frac{1}{2}\int_{-1}^{1}(5x^4 - 3x^2)\,dx.\end{aligned}$$

Both powers are even, so:

$$\begin{aligned}\langle P_1, P_3\rangle &= \int_0^1 (5x^4 - 3x^2)\,dx \\ &= \left[x^5 - x^3\right]_0^1 \\ &= (1-1) - (0-0) \\ &= 0.\end{aligned}$$

Therefore P_1 and P_3 are orthogonal.

Why The Chebyshev Weight Looks Singular

Chebyshev orthogonality uses:

$$\int_{-1}^{1} \frac{T_m(x)T_n(x)}{\sqrt{1-x^2}}\,dx.$$

Although the weight becomes unbounded at $x = \pm 1$, the singularities are integrable. Use $x = \cos\theta$:

$$dx = -\sin\theta\,d\theta, \qquad \sqrt{1-x^2} = \sin\theta$$

for $0 < \theta < \pi$. The weight and differential cancel:

$$\frac{dx}{\sqrt{1-x^2}} = -d\theta.$$

After reversing the transformed limits:

$$\int_{-1}^{1} \frac{T_m(x)T_n(x)}{\sqrt{1-x^2}}\,dx = \int_0^{\pi} \cos(m\theta)\cos(n\theta)\,d\theta.$$

For $m \neq n$, the right side is zero by trigonometric orthogonality. The unusual Chebyshev weight is precisely what converts the x-integral into a standard cosine inner product.

Block 7 Summary

Self-adjoint form turns two differential equations with different parameter values into an integral orthogonality relation. Each polynomial family carries the interval and weight naturally associated with its equation.

Expanding Functions In An Orthogonal Basis

The Goal Of This Block

The goal is to use orthogonality to determine expansion coefficients and show why these polynomials are more than isolated exact solutions.

The Projection Formula

Suppose orthogonal functions $\phi_0, \phi_1, \ldots$ are used to represent:

$$f(x) = \sum_{n=0}^{N} c_n \phi_n(x).$$

Take the weighted inner product with ϕ_m:

$$\langle f, \phi_m \rangle_w = \sum_{n=0}^{N} c_n \langle \phi_n, \phi_m \rangle_w.$$

Orthogonality makes every term with $n \neq m$ vanish:

$$\langle f, \phi_m \rangle_w = c_m \langle \phi_m, \phi_m \rangle_w.$$

Divide by the nonzero squared norm:

$$\boxed{c_m = \frac{\langle f, \phi_m \rangle_w}{\langle \phi_m, \phi_m \rangle_w}}.$$

This is the continuous analogue of finding a vector component by projecting onto an orthogonal direction.

Worked Legendre Expansion Of X Cubed

Represent $f(x) = x^3$ using the odd Legendre polynomials P_1 and P_3:

$$x^3 = c_1 P_1(x) + c_3 P_3(x).$$

First calculate c_1. Since $P_1 = x$:

$$\begin{aligned}\langle x^3, P_1\rangle &= \int_{-1}^{1} x^3(x)\,dx \\ &= \int_{-1}^{1} x^4\,dx \\ &= \frac{2}{5}.\end{aligned}$$

The squared norm is:

$$\begin{aligned}\langle P_1, P_1\rangle &= \int_{-1}^{1} x^2\,dx \\ &= \frac{2}{3}.\end{aligned}$$

Therefore:

$$\begin{aligned}c_1 &= \frac{2/5}{2/3} \\ &= \frac{3}{5}.\end{aligned}$$

Now calculate c_3. Using $P_3 = (5x^3 - 3x)/2$:

$$\begin{aligned}\langle x^3, P_3\rangle &= \frac{1}{2}\int_{-1}^{1} (5x^6 - 3x^4)\,dx \\ &= \frac{1}{2}\left(5 \cdot \frac{2}{7} - 3 \cdot \frac{2}{5}\right) \\ &= \frac{1}{2}\left(\frac{10}{7} - \frac{6}{5}\right) \\ &= \frac{4}{35}.\end{aligned}$$

Calculate the squared norm directly:

$$\begin{aligned}\langle P_3, P_3\rangle &= \frac{1}{4}\int_{-1}^{1}(25x^6 - 30x^4 + 9x^2)\,dx \\ &= \frac{1}{4}\left(25\cdot\frac{2}{7} - 30\cdot\frac{2}{5} + 9\cdot\frac{2}{3}\right) \\ &= \frac{1}{4}\left(\frac{50}{7} - 12 + 6\right) \\ &= \frac{2}{7}.\end{aligned}$$

Hence:

$$\begin{aligned}c_3 &= \frac{4/35}{2/7} \\ &= \frac{2}{5}.\end{aligned}$$

The expansion is therefore:

$$\boxed{x^3 = \frac{3}{5}P_1(x) + \frac{2}{5}P_3(x)}.$$

Check The Expansion Algebraically

Insert the explicit polynomials:

$$\begin{aligned}\frac{3}{5}P_1 + \frac{2}{5}P_3 &= \frac{3}{5}x + \frac{2}{5}\left[\frac{1}{2}(5x^3 - 3x)\right] \\ &= \frac{3}{5}x + \frac{1}{5}(5x^3 - 3x) \\ &= \frac{3}{5}x + x^3 - \frac{3}{5}x \\ &= x^3.\end{aligned}$$

The projection calculation and direct algebra agree.

Exact Expansion Versus Approximation

The previous expansion is exact because x^3 lies in the span of P_1 and P_3. For a non-polynomial function, a finite sum generally gives an approximation:

$$f(x) \approx \sum_{n=0}^{N} c_n P_n(x).$$

The projection coefficients choose the approximation whose weighted squared error is smallest within the selected polynomial span. Increasing N adds more orthogonal directions and can capture more detail.

Block 8 Summary

Orthogonality isolates one expansion coefficient at a time. The same polynomials that solve classical differential equations therefore provide natural bases for representing and approximating other functions.

A Reliable Classical-Polynomial Workflow

The Goal Of This Block

The goal is to organise recognition, construction, normalization, and verification into a dependable sequence.

The Workflow

Step 1: Identify the family. Compare every coefficient with the standard differential equation. Do not identify a family from one coefficient alone.

Step 2: Recover the degree parameter. Solve the parameter relation. For example, a Legendre coefficient of 20 requires $n(n+1) = 20$, whereas a Hermite coefficient of 10 requires $2n = 10$.

Step 3: Use a separate series index. If n labels the polynomial degree, use k for the power-series index:

$$y = \sum_{k=0}^{\infty} a_k x^k.$$

Step 4: Derive the recurrence locally. Substitute y, y', and y'', align the powers, and show which coefficient equation comes from x^k.

Step 5: Locate the terminating factor. Find the exact index at which a numerator such as $k - n$ vanishes.

Step 6: Select the terminating chain. For a two-step recurrence, retain the parity chain matching n and set the other starting coefficient to zero. For Laguerre's one-step recurrence, begin from a_0.

Step 7: Apply the stated normalization. Use $P_n(1) = 1$, $T_n(1) = 1$, the leading coefficient 2^n for H_n, or $L_n(0) = 1$.

Step 8: Verify the result. Check the degree, normalization, and differential equation. If orthogonality is claimed, include the correct interval and weight.

Step 9: Choose an efficient secondary tool. Use a neighbour recurrence for later members, a Rodrigues formula for one explicit member, or a generating function when several low-order members are needed together.

Common Mistake: Confusing N And K

In:

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0,$$

n is fixed while one chosen Legendre equation is being solved. In:

$$y = \sum_{k=0}^{\infty} a_k x^k,$$

k runs through all coefficient indices. Reusing n for both jobs can make the termination step impossible to read.

Common Mistake: Stopping At One Zero Coefficient

Showing $a_{n+2} = 0$ is not enough by itself. Use the recurrence once more:

$$a_{n+4} = R(n+2)a_{n+2} = 0,$$

and similarly for every later coefficient in that chain. The recurrence, not wishful truncation, proves that the series has terminated.

Common Mistake: Mixing Normalizations

The equation is homogeneous, so a nonzero multiple still solves it. However, the name L_2 in this chapter means the normalization $L_2(0) = 1$:

$$L_2(x) = 1 - 2x + \frac{1}{2}x^2.$$

The polynomial $2 - 4x + x^2$ is also a solution, but it is $2L_2$, not the normalized L_2 itself.

Common Mistake: Omitting The Weight

Chebyshev orthogonality does not mean:

$$\int_{-1}^{1} T_m(x)T_n(x)\,dx = 0$$

for every $m \neq n$. The correct inner product contains:

$$w(x) = \frac{1}{\sqrt{1-x^2}}.$$

Orthogonality is always a statement about a family, an interval, and a weight together.

Common Mistake: Reading A Generating Coefficient Too Quickly

First inspect the right side of the generating identity. In:

$$e^{2xt-t^2} = \sum_{n=0}^{\infty} H_n(x)\frac{t^n}{n!},$$

the coefficient of t^n must be multiplied by $n!$ to recover H_n.

Block 9 Summary

Recognise the full equation, separate the degree label from the series index, prove termination through the recurrence, normalize explicitly, and attach the correct weight and interval to every orthogonality claim.

Original Practice Set

Problems 1 To 4: Recognition And Basic Construction

Problem 1: Identify Four Families. Identify the polynomial family and degree n associated with each equation.

$$\begin{aligned} &\text{(a)} \quad (1-x^2)y'' - xy' + 49y = 0, \\ &\text{(b)} \quad y'' - 2xy' + 12y = 0, \\ &\text{(c)} \quad xy'' + (1-x)y' + 4y = 0, \\ &\text{(d)} \quad (1-x^2)y'' - 2xy' + 30y = 0. \end{aligned}$$

Problem 2: Rebuild The Chebyshev Recurrence. Starting from:

$$(1-x^2)y'' - xy' + n^2 y = 0,$$

derive the recurrence relating a_{k+2} to a_k.

Problem 3: Construct P Two. Use the Legendre coefficient recurrence and the normalization $P_2(1) = 1$ to construct $P_2(x)$.

Problem 4: Generate T Three. Use:

$$T_{n+1} = 2xT_n - T_{n-1},$$

together with $T_1 = x$ and $T_2 = 2x^2 - 1$, to find T_3.

Problems 5 To 8: Series, Rodrigues, And Verification

Problem 5: Construct H Three By Series. Use the Hermite recurrence and the physicists' leading-coefficient normalization to construct $H_3(x)$.

Problem 6: Construct L Two By Series. Use the Laguerre recurrence and $L_2(0) = 1$ to construct $L_2(x)$.

Problem 7: Use Rodrigues For P Three. Starting from:

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n,$$

calculate $P_3(x)$.

Problem 8: Verify A Laguerre Polynomial. Verify directly that:

$$L_2(x) = 1 - 2x + \frac{1}{2}x^2$$

solves the $n = 2$ Laguerre equation.

Problems 9 To 12: Orthogonality, Projection, And Scale

Problem 9: Check Legendre Orthogonality. Verify directly that $P_0(x) = 1$ and $P_2(x) = (3x^2 - 1)/2$ are orthogonal on $[-1, 1]$ with weight 1.

Problem 10: Transform A Chebyshev Inner Product. Use $x = \cos\theta$ to show that the weighted inner product of T_1 and T_2 on $[-1, 1]$ becomes:

$$\int_0^\pi \cos\theta \cos(2\theta)\, d\theta,$$

and evaluate it.

Problem 11: Expand X Squared In Legendre Polynomials. Use weighted projections to find c_0 and c_2 in:

$$x^2 = c_0 P_0(x) + c_2 P_2(x).$$

Problem 12: Separate A Solution From Its Normalization. Let:

$$q(x) = x^2 - 4x + 2.$$

Show that q solves the $n = 2$ Laguerre equation. Then determine its relation to the normalized polynomial L_2.

Worked Answers: Problems 1 To 4

Answer 1.

Equation (a). Compare:

$$(1 - x^2)y'' - xy' + 49y = 0$$

with the Chebyshev equation:

$$(1 - x^2)y'' - xy' + n^2y = 0.$$

Thus:

$$n^2 = 49.$$

Since the family uses $n \geq 0$:

$$\boxed{n = 7, \text{ so the polynomial is } T_7.}$$

Equation (b). Compare:

$$y'' - 2xy' + 12y = 0$$

with the Hermite equation:

$$y'' - 2xy' + 2ny = 0.$$

Therefore:

$$2n = 12,$$

so:

$$\boxed{n = 6, \text{ giving } H_6.}$$

Equation (c). The equation:

$$xy'' + (1 - x)y' + 4y = 0$$

has Laguerre form with $n = 4$. Hence:

$$\boxed{L_4.}$$

Equation (d). Compare:

$$(1-x^2)y'' - 2xy' + 30y = 0$$

with the Legendre equation. The parameter satisfies:

$$n(n+1) = 30.$$

Move all terms to one side:

$$n^2 + n - 30 = 0.$$

Factor:

$$(n-5)(n+6) = 0.$$

The nonnegative choice is $n = 5$, so:

$$\boxed{P_5.}$$

Answer 2. Use:

$$y = \sum_{k=0}^{\infty} a_k x^k.$$

The required series forms are:

$$\begin{aligned} y'' &= \sum_{k=0}^{\infty} (k+2)(k+1)a_{k+2}x^k, \\ x^2y'' &= \sum_{k=0}^{\infty} k(k-1)a_k x^k, \\ xy' &= \sum_{k=0}^{\infty} k a_k x^k, \\ n^2 y &= \sum_{k=0}^{\infty} n^2 a_k x^k. \end{aligned}$$

Substitute into the expanded equation:

$$y'' - x^2y'' - xy' + n^2y = 0.$$

The coefficient of x^k gives:

$$(k+2)(k+1)a_{k+2} + [n^2 - k(k-1) - k]a_k = 0.$$

Since $k(k-1) + k = k^2$:

$$(k+2)(k+1)a_{k+2} + (n^2 - k^2)a_k = 0.$$

Move the term containing a_k to the right side:

$$\begin{aligned}(k+2)(k+1)a_{k+2} &= -(n^2-k^2)a_k \\ &= (k^2-n^2)a_k.\end{aligned}$$

Now divide both sides by $(k+2)(k+1)$:

$$\boxed{a_{k+2} = \frac{k^2-n^2}{(k+2)(k+1)}a_k}.$$

Answer 3. For $n=2$, the Legendre recurrence is:

$$a_{k+2} = \frac{(k-2)(k+3)}{(k+2)(k+1)}a_k.$$

Use the even chain. At $k=0$:

$$\begin{aligned}a_2 &= \frac{(0-2)(0+3)}{(2)(1)}a_0 \\ &= -3a_0.\end{aligned}$$

Using $a_2 = -3a_0$, continue with $k=2$:

$$a_4 = 0.$$

Thus:

$$y = a_0(1-3x^2).$$

Apply $P_2(1) = 1$:

$$1 = a_0(1-3) = -2a_0.$$

Therefore $a_0 = -1/2$, and:

$$\boxed{P_2(x) = \frac{1}{2}(3x^2 - 1)}.$$

Answer 4. Set $n = 2$ in the neighbour recurrence:

$$T_3 = 2xT_2 - T_1.$$

Substitute $T_2 = 2x^2 - 1$ and $T_1 = x$:

$$\begin{aligned} T_3 &= 2x(2x^2 - 1) - x \\ &= 4x^3 - 2x - x \\ &= 4x^3 - 3x. \end{aligned}$$

Hence:

$$\boxed{T_3(x) = 4x^3 - 3x}.$$

Worked Answers: Problems 5 To 8

Answer 5. For $n = 3$, the Hermite recurrence becomes:

$$a_{k+2} = \frac{2(k-3)}{(k+2)(k+1)} a_k.$$

Use the odd chain and set $a_0 = 0$. At $k = 1$:

$$\begin{aligned} a_3 &= \frac{2(1-3)}{(3)(2)} a_1 \\ &= -\frac{2}{3} a_1. \end{aligned}$$

At $k = 3$, the terminating factor appears:

$$a_5 = 0.$$

The polynomial therefore has the form:

$$y = a_1 x - \frac{2}{3} a_1 x^3.$$

The physicists' normalization requires the leading coefficient to be $2^3 = 8$. Hence:

$$-\frac{2}{3} a_1 = 8.$$

Multiply by $-3/2$:

$$a_1 = -12.$$

Substitute this value:

$$\boxed{H_3(x) = 8x^3 - 12x}.$$

Answer 6. For $n = 2$, the Laguerre recurrence is:

$$a_{k+1} = \frac{k-2}{(k+1)^2} a_k.$$

The normalization $L_2(0) = 1$ gives $a_0 = 1$. At $k = 0$:

$$a_1 = \frac{-2}{1} a_0 = -2.$$

At $k = 1$:

$$\begin{aligned} a_2 &= \frac{1-2}{(2)^2} a_1 \\ &= -\frac{1}{4}(-2) \\ &= \frac{1}{2}. \end{aligned}$$

At $k = 2$:

$$a_3 = 0.$$

Thus:

$$\boxed{L_2(x) = 1 - 2x + \frac{1}{2}x^2}.$$

Answer 7. For $n = 3$, Rodrigues' formula gives:

$$P_3(x) = \frac{1}{2^3(3!)} \frac{d^3}{dx^3}(x^2 - 1)^3.$$

First expand the cube:

$$(x^2 - 1)^3 = x^6 - 3x^4 + 3x^2 - 1.$$

Differentiate once:

$$6x^5 - 12x^3 + 6x.$$

Differentiate a second time:

$$30x^4 - 36x^2 + 6.$$

Differentiate a third time:

$$120x^3 - 72x.$$

The denominator is:

$$2^3(3!) = 8 \cdot 6 = 48.$$

Therefore:

$$\begin{aligned} P_3(x) &= \frac{120x^3 - 72x}{48} \\ &= \frac{1}{2}(5x^3 - 3x). \end{aligned}$$

Answer 8. For $n = 2$, the Laguerre equation is:

$$xy'' + (1 - x)y' + 2y = 0.$$

Let:

$$y = 1 - 2x + \frac{1}{2}x^2.$$

Its derivatives are:

$$y' = -2 + x, \qquad y'' = 1.$$

Evaluate each equation term:

$$xy'' = x,$$

$$\begin{aligned}(1 - x)y' &= (1 - x)(x - 2) \\ &= x - 2 - x^2 + 2x \\ &= -x^2 + 3x - 2,\end{aligned}$$

and:

$$2y = 2 - 4x + x^2.$$

Add them:

$$\begin{aligned}xy'' + (1 - x)y' + 2y &= x + (-x^2 + 3x - 2) + (2 - 4x + x^2) \\ &= 0.\end{aligned}$$

Thus the proposed L_2 satisfies the equation.

Worked Answers: Problems 9 To 12

Answer 9. Use:

$$P_0(x) = 1, \qquad P_2(x) = \frac{1}{2}(3x^2 - 1).$$

Their inner product with weight 1 is:

$$\begin{aligned}\langle P_0, P_2\rangle &= \int_{-1}^{1} 1 \cdot \frac{1}{2}(3x^2 - 1)\,dx \\ &= \frac{1}{2}\left(3\int_{-1}^{1} x^2\,dx - \int_{-1}^{1} 1\,dx\right).\end{aligned}$$

Evaluate the two integrals:

$$\int_{-1}^{1} x^2\,dx = \frac{2}{3}, \qquad \int_{-1}^{1} 1\,dx = 2.$$

Substitute:

$$\begin{aligned}\langle P_0, P_2\rangle &= \frac{1}{2}\left(3 \cdot \frac{2}{3} - 2\right) \\ &= \frac{1}{2}(2 - 2) \\ &= 0.\end{aligned}$$

Therefore P_0 and P_2 are orthogonal on $[-1, 1]$.

Answer 10. The Chebyshev inner product is:

$$\int_{-1}^{1} \frac{T_1(x)T_2(x)}{\sqrt{1-x^2}}\,dx.$$

Set $x = \cos\theta$. Then:

$$dx = -\sin\theta\,d\theta, \qquad \sqrt{1-x^2} = \sin\theta.$$

Also:

$$T_1(\cos\theta) = \cos\theta, \qquad T_2(\cos\theta) = \cos(2\theta).$$

As x increases from -1 to 1, θ decreases from π to 0. Therefore:

$$\begin{aligned}\int_{-1}^{1} \frac{T_1T_2}{\sqrt{1-x^2}}\,dx &= \int_{\pi}^{0} \frac{\cos\theta\cos(2\theta)}{\sin\theta}(-\sin\theta)\,d\theta \\ &= \int_{0}^{\pi} \cos\theta\cos(2\theta)\,d\theta.\end{aligned}$$

Use the product-to-sum identity:

$$\cos\theta\cos(2\theta) = \frac{1}{2}[\cos\theta + \cos(3\theta)].$$

Hence:

$$\begin{aligned}\int_{0}^{\pi} \cos\theta\cos(2\theta)\,d\theta &= \frac{1}{2}\int_{0}^{\pi}[\cos\theta + \cos(3\theta)]\,d\theta \\ &= \frac{1}{2}\left[\sin\theta + \frac{1}{3}\sin(3\theta)\right]_{0}^{\pi} \\ &= 0.\end{aligned}$$

Answer 11. Seek:

$$x^2 = c_0 P_0 + c_2 P_2.$$

For c_0, use the projection formula:

$$c_0 = \frac{\langle x^2, P_0 \rangle}{\langle P_0, P_0 \rangle}.$$

Calculate the numerator and denominator:

$$\langle x^2, P_0 \rangle = \int_{-1}^{1} x^2 \, dx = \frac{2}{3},$$

$$\langle P_0, P_0 \rangle = \int_{-1}^{1} 1 \, dx = 2.$$

Therefore:

$$c_0 = \frac{2/3}{2} = \frac{1}{3}.$$

For c_2:

$$c_2 = \frac{\langle x^2, P_2 \rangle}{\langle P_2, P_2 \rangle}.$$

The numerator is:

$$\begin{aligned} \langle x^2, P_2 \rangle &= \frac{1}{2} \int_{-1}^{1} (3x^4 - x^2) \, dx \\ &= \frac{1}{2} \left(3 \cdot \frac{2}{5} - \frac{2}{3} \right) \\ &= \frac{4}{15}. \end{aligned}$$

The denominator is:

$$\begin{aligned}\langle P_2, P_2 \rangle &= \frac{1}{4} \int_{-1}^{1} (9x^4 - 6x^2 + 1)\, dx \\ &= \frac{1}{4} \left(9 \cdot \frac{2}{5} - 6 \cdot \frac{2}{3} + 2 \right) \\ &= \frac{2}{5}.\end{aligned}$$

Thus:

$$c_2 = \frac{4/15}{2/5} = \frac{2}{3}.$$

The expansion is:

$$\boxed{x^2 = \frac{1}{3} P_0(x) + \frac{2}{3} P_2(x)}.$$

Answer 12. For $n = 2$, the Laguerre equation is:

$$xy'' + (1 - x)y' + 2y = 0.$$

Let:

$$q = x^2 - 4x + 2.$$

Differentiate:

$$q' = 2x - 4, \qquad q'' = 2.$$

Substitute:

$$xq'' + (1 - x)q' + 2q = 2x + (1 - x)(2x - 4) + 2(x^2 - 4x + 2).$$

Expand the product:

$$\begin{aligned}(1 - x)(2x - 4) &= 2x - 4 - 2x^2 + 4x \\ &= -2x^2 + 6x - 4.\end{aligned}$$

Now combine all terms:

$$\begin{aligned}xq'' + (1 - x)q' + 2q &= 2x + (-2x^2 + 6x - 4) + (2x^2 - 8x + 4) \\ &= 0.\end{aligned}$$

Thus q is a solution. Compare it with:

$$L_2(x) = 1 - 2x + \frac{1}{2}x^2.$$

Multiplying L_2 by 2 gives:

$$2L_2(x) = 2 - 4x + x^2 = q(x).$$

Therefore:

$$\boxed{q = 2L_2}.$$

It solves the equation but is not the normalized L_2, because $q(0) = 2$ rather than 1.

Chapter 29 Final Summary

The four classical equations select polynomial solutions when the degree parameter is a nonnegative integer:

$$
\begin{aligned}
(1-x^2)y'' - 2xy' + n(n+1)y = 0 &\quad \longrightarrow P_n,\\
(1-x^2)y'' - xy' + n^2 y = 0 &\quad \longrightarrow T_n,\\
y'' - 2xy' + 2ny = 0 &\quad \longrightarrow H_n,\\
xy'' + (1-x)y' + ny = 0 &\quad \longrightarrow L_n.
\end{aligned}
$$

Their common construction pattern is:

$$
\boxed{
\begin{aligned}
\text{identify the equation} &\longrightarrow \text{derive the coefficient recurrence}\\
&\longrightarrow \text{locate its terminating factor}\\
&\longrightarrow \text{select the terminating chain}\\
&\longrightarrow \text{apply the family normalization}\\
&\longrightarrow \text{verify the equation and degree.}
\end{aligned}
}
$$

Self-adjoint form supplies the appropriate weighted orthogonality. That orthogonality then turns each family into a basis for exact expansions and approximations.

a classical polynomial is simultaneously a terminated series solution, a normalized member of a recurrence family, and an orthogonal basis function.